

Michel Domenella

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ABOUT ME

Human-centered product engineer with two years of medical device experience, from neonatal ventilators to at-home hemodialysis, now building formal design expertise through a Master's at Northwestern. I'm drawn to the space where patient needs and engineering constraints meet, and I thrive moving fluidly between field research and the design team. I believe that great medical products come from teams that treat user understanding as just as rigorous a discipline as the engineering itself.

CERTIFICATIONS

International Fluid Power Society
Pneumatic Specialist

June 2024

Certified SolidWorks Associate
Mechanical Design

June 2021

SKILLS

User Research Methods

- Contextual Inquiry
- Stakeholder Interviews
- Usability Testing
- Synthesis and Framing

Regulatory Experience

- RoHS 3
- FDA Device Classification
- 510(k) Pathways

Engineering Skills

- CAD
- Machining
- 3D Printing
- Validation Testing
- MatLab

EDUCATION

MS	Northwestern University , Engineering Design Innovation	Aug 2025 - Mar 2027
BS	University of Notre Dame , Mechanical Engineering	Aug 2019 - May 2023

EXPERIENCE

Clippard , New Product Development Engineer	Jul 2023 - Jun 2025 Cincinnati, OH
• Designed a panel-mount pinch valve for a cell and gene therapy client, iterating on fastener design through internal usability testing to accommodate multi-modal tightening and incorporating a gasket seal to meet wipe-down cleanliness requirements • Resolved a critical failure for an at-home hemodialysis valve by closely collaborating with the client's engineering team — performing root-cause analysis that brought the beta failure rate to 0% • Designed and validated a RoHS 3-compliant oxygen mixing blender for a neonatal open care warmer ventilator, delivering a fully functional beta unit within 3 months	
Northwestern , Needfinding in Medical Environments	April 2026 - Present Chicago, IL
• Conducted clinical observations and stakeholder interviews across a Northwestern GI endoscopy clinic — including physicians, PAs, and nurses — to surface unmet needs in Barrett's Esophagus screening workflows • Synthesized findings into journey maps, process maps, and opportunity frameworks to identify gaps between patient experience and clinical operational constraints • Developed "How Might We" statements and engineering problem statements bridging patient tolerability, clinician workflow, and regulatory viability to prioritize actionable design opportunities	
Northwestern , P&G Sponsored Design Project	Sept 2025 - Dec 2025 Chicago, IL
• Led a multidisciplinary team of 5 engineers, designers, and a cognitive scientist through a full HCD process, from in-home and central site consumer research sessions to a fully functioning physical prototype, for a P&G sponsored packaging project • Synthesized findings from 5 consumer research sessions into actionable design requirements, directly informing iterative prototype development presented to P&G stakeholders	